

BFS MAZE SOLVER: UNINFORMED SEARCH REPORT

Formal Demonstration of Search Optimality on Uniform-Cost Grids

1. PROBLEM STATEMENT & OBJECTIVES

This project implements an uninformed search agent using Breadth-First Search (BFS) to find the strictly shortest (optimal) path in 2D grid mazes with arbitrary obstacles.

Key requirements fulfilled:	Maze Preset	Dimensions	BFS (Optimal)	DFS (Baseline)	BFS Extra Steps	Suboptimality
	* 2D Grid & Matrix Input: Support for ASCII string parsing, 2D boolean/integer matrices, and processing.					
	* Obstacle & Boundary Collision: Robust handling of out-of-bounds cells, wall blocks, and disconnections.					
	* Visualisation: Color-coded path reconstruction, exploration history, and distance layer contours.					
	* Formal & Empirical Optimality: Demonstration of BFS optimality on uniform-cost graphs compared to DFS.					
2. THEORETICAL FOUNDATION: WHY BFS IS OPTIMAL ON UNIFORM-COST GRIDS	Branching Factor	6x7	12	16	4	1.33x longer
In a grid where orthogonal moves (Up, Down, Left, Right) each possess an identical unit cost $c(e) = 1$.	BFS Tree	5x11	4	22	+28	8.0x longer!
1. Monotonic Queue Property: BFS maintains a First-In, First-Out (FIFO) queue. Nodes are dequeued in strictly non-decreasing order of cost $g(n)$ from start:	$d(\text{Start}, v_1) \leq d(\text{Start}, v_2) \leq \dots \leq d(\text{Start}, v_k)$					
2. Queue Invariant: At any instant during execution, the queue contains vertices with distance at least d^* .						
3. First-Visit Optimality: When the Goal node G is first dequeued at distance d^* , any node remaining in the frontier or unvisited states has cost $\geq d^*$. Therefore, no alternate path through unexplored states possibly have fewer edges. Hence, BFS is guaranteed to be optimal.						

3. EMPIRICAL BENCHMARK SUMMARY TABLE

Below is the empirical comparison of path length (number of moves) between BFS and DFS:

EMPIRICAL BENCHMARKS & WAVEFRONT VISUALIZATIONS

